Class Meeting #15: Pre-Meeting Activity

Why don't we see *ortho* subtituents in Hammett's substituent parameters? Let's graph some data and consider the results. This is an assigned problem to be completed before meeting #15. Bring a printed copy of your plot.

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Including ortho Substituents

It is easy to measure the electronic effect of *ortho* substituents on the acid equilibrium of benzoic acids and from that data calculate a set of σ values for these substituents. If you examine the Williams data set of substituent parameters[1] you will see that there are σ values for a collection of *ortho* substituents present.

In table 1, the rates for base hydrolysis of various ethyl benzoates[2,3] are presented. Construct a Hammett plot and comment on the quality of the plot when the *ortho* substituents are included and when they are not. Bring a copy of your plot to the class meeting for show-and-tell.

[1] See the `LFER-Williams.csv` file available in the data repository for the course at https://raw.githubusercontent.com/blinkletter/LFER-QSAR/refs/heads/main/data/LFER_Williams.csv.

[2] "Kinetic Studies of the Hydrolysis of Esters. I. Alkaline Hydrolysis of Esters of Substituted Benzoic Acids." E. Tommila, *Ann. Acad. Sci. Fennicae*, **1941**, Ser. A 57, 3.

Substituent	σ	k_{OH} /10^{-3} $M^{-1}s^{-1}$
m-OCH_3	0.12	3.92
p-$N(CH_3)_2$	−0.83	0.0634
p-Cl	0.23	11.7
m-Cl	0.37	18.2
p-Br	0.23	13.9
m-Br	0.39	17.9
p-I	0.18	12.2
m-I	0.35	15.0
p-F	0.06	5.86
p-CN	0.66	157
m-CN	0.56	103
m-NO_2	0.72	137
p-NO_2	0.78	246
p-NH_2	−0.66	0.0864
m-NH_2	−0.16	1.66
H	0	2.89
m-CH_3	−0.07	1.69
p-CH_3	−0.17	1.14
o-CN	1.06	122
o-OEt	−0.01	1.16
o-CH_3	0.29	0.338
o-Cl	1.26	4.40
o-NO_2	2.03	16.9
o-Br	1.35	3.00
o-I	1.34	1.63

Table 1: Rates of base hydrolysis for various ethyl benzoates at 25 °C.[2,3]
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You can download this data set from https://raw.githubusercontent.com/blinkletter/4410PythonNotebooks/refs/heads/main/Class_17/data/4410_15-PreClassActivityData.csv.

[3] I could not find the Tommila paper, but our librarian did. I then obtained a copy through interlibrary loan.

This data is available in other collected works and reviews throughout the literature. For example, you can find this data in "Free Energy Relationships in Organic and Bio-organic Chemistry" by Andrew Williams, *The Royal Society of Chemistry, Cambridge*, **2003**, pp 46. https://doi.org/10.1039/9781847550927. We have access to the E-textbook version via the UPEI library.

Williams reported the values calculated from Arrhenius plots of the kinetics and not the experimental value at 25 °C. This was a way to include the data at other temperatures in the final reported value. Using the raw data from the paper, we could construct an Eyring plot for each result in table 1 and then calculate our own values. We would also be able to calculate the standard error for the results. Would we observe an isokinetic temperature? Would the errors be significant? This could be the start of an exploration...

When you're serious, always seek out the source. If you want to read the original paper, just ask me for a copy.